

Paria Zandihamedani

Machine Learning & AI Engineer | [Data Science](#)

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Profile

MSc student in Electronic Engineering with experience in machine learning, deep learning, data science, and AI applications. Worked on projects in time-series forecasting, medical imaging, data engineering, computer vision, and retrieval-augmented generation using Python and modern ML tools. Currently completing an MSc thesis focused on personalized modeling, individual baseline analysis, trajectory modeling, and future-state prediction from longitudinal health data.

Professional Experience

Machine Learning Thesis Intern MIA Health Trondheim, Norway	2026 – Present
- Working on personalized modeling and prediction from longitudinal health data as part of an MSc thesis collaboration. - Developing methods for individual baseline estimation, trajectory analysis, and prediction of future personalized states. - Exploring structured representations of individual health state for use in digital health and AI-assisted applications.	
AI Developer RAGdoll Project, IMTEL Trondheim, Norway	2025 – 2026
- Worked in an 8-member Scrum team on a customer-driven educational assistant based on retrieval-augmented generation. - Contributed to a system using document retrieval, embeddings, contextual question answering, FastAPI, Git, and iterative client feedback.	
Data Engineer Behin Pardazeshgaran Mad Shahr Iran	2021 – 2023
- Managed and validated patient data across electronic hospital systems, including correction of inconsistent records and support for appointment, billing, and payment workflows. - Worked with Python and SQL for data processing, transformation, reporting, and operational monitoring. - Used data platforms including ClickHouse and BigQuery for healthcare data workflows.	

Education

MSc in Electronic Engineering – Exchange Student Norwegian University of Science and Technology (NTNU) Trondheim, Norway	2025 – Present
MSc Thesis: Personalized Health Modeling and AI Agent Support	
MSc in Electronic Engineering University of Bologna Bologna, Italy	2024 – Present
Track: Electronics for Intelligent Systems, Big Data and Internet of Things	
BSc in Electrical Engineering Hamedan University of Technology Iran	2019 – 2024
Bachelor's Thesis: Comparative Study of Classical and Deep Learning-Based Medical Image Fusion Methods	

Selected Projects

Industrial Raw Material Delivery Forecasting	Machine Learning / Time-Series
- Developed a cumulative forecasting model for Hydro ASA to predict raw-material deliveries across future forecast horizons. - Combined a 90-day historical delivery baseline with LightGBM quantile regression to model residual corrections. - Engineered 14 features from recent delivery statistics, trend, forecast horizon, and purchase-order information. - Reduced P20 validation quantile loss by 35.8% compared with the strongest baseline.	
GitHub repository	
Breast Cancer DCE-MRI Classification	Deep Learning / Medical Imaging
- Developed 3D ResNet18 and 2D MIP ResNet50 pipelines for breast cancer classification from multi-centre DCE-MRI data. - Applied image preprocessing, augmentation, class weighting, test-time augmentation, and logit-space model ensembling. - Achieved approximately 0.78 AUROC and ranked 5th of 28 teams in the ODELIA 2025 challenge.	
GitHub repository	
Porto Taxi Trajectory Analysis	Data Engineering / SQL / Geospatial
Built a Python/MySQL pipeline to clean, restructure, and load 1.67M taxi trips and 83.4M GPS points into relational tables.	

- Used SQL window functions and spatial queries for trip duration, idle-time, proximity, and location analysis.
- Improved a spatial query from approximately **8 minutes to 23 seconds** using indexing and grid filtering.

Snow Pole Detection for Winter Road Perception

Computer Vision / Object Detection

- Developed a one-class object-detection pipeline for snow poles in winter road scenes from Trøndelag.
- Trained and compared YOLO and RT-DETR models on NTNU’s IDUN HPC environment using SLURM.
- Evaluated models across different image datasets and observed consistently stronger performance from RT-DETR in the experiments.

SkyBite Autonomous Drone Delivery System

Python / IoT / Distributed Systems

- Contributed to the architecture of the communication flow for an autonomous drone-delivery system.
- Implemented fleet-management logic for drone coordination, mission states, and delivery updates.
- Integrated a Python backend with MQTT/Mosquitto, Raspberry Pi-oriented hardware, and user-facing applications.

[GitHub repository](#)

Neural Jigsaw Image Reconstruction

Deep Learning / Computer Vision

- Built models to reconstruct 96×96 RGB images from nine shuffled image patches.
- Compared CNN, U-Net, and Transformer architectures for image arrangement and reconstruction.
- Evaluated reconstruction quality using MAE and SSIM.

[GitHub repository](#)

Technical Skills

Programming

Python, SQL, C, C++

Machine Learning

PyTorch, TensorFlow, scikit-learn, LightGBM, XGBoost

ML Areas

Time-Series Forecasting, Computer Vision, Object Detection

Generative AI

RAG, Embeddings, Semantic Retrieval, FastAPI, LLM Applications

Data Engineering & Analysis

ETL, Data Modeling, MySQL, BigQuery, ClickHouse, MongoDB, pandas, NumPy, Power BI

Tools

Docker, Git, SLURM

Languages

Persian: Native

English: Professional Working Proficiency